

Sabyasachi Das

Senior Data Engineer

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SUMMARY

Senior Data Engineer with **6 years of experience** designing and delivering scalable data pipelines, lakehouse architectures, and cloud-native solutions. Proficient in **Spark, Databricks, and AWS**, with hands-on **GenAI** experience building **text-to-SQL solutions** using RAG and LLM APIs. Adept at working across pharma, BFSI, and retail domains in Agile environments.

SKILLS

Languages	Scala, Python, Java, Shell Scripting
Big Data	Spark (Scala / PySpark), Hive, Kafka, Spark Streaming, Databricks, Data Modelling
Cloud / AWS	S3, Glue, Athena, RDS, Redshift, DynamoDB, LocalStack
Databases	MySQL, DynamoDB
Orchestration	Apache Airflow, Databricks Workflow
GenAI	RAG, Evals, LLM API / Tool Calling, Ollama, HuggingFace
DevOps	Git, Maven, Jenkins, CI/CD

PROFESSIONAL EXPERIENCE

Thoughtworks

July 2023 – Present

Senior Data/GenAI Engineer

Bangalore, India

- Built a **data model from scratch** for daily ingestion of **800GB-1TB** of data, for a pharma client. Implemented it in **Databricks** using **medallion architecture**, providing a unified view of all exceptions and errors in medicine delivery across integrated data lake sources.
- Designed and scheduled **Databricks Workflow** jobs for daily batch runs; extended the platform with a **GenAI text-to-SQL** feature using RAG and LLM APIs, enabling non-technical users to query data via natural language.
- Led a data model redesign for a footwear client, migrating legacy **OracleDB (SQL)** to **DynamoDB (NoSQL)**; followed **AI First Development** best practices and validated the solution using **LocalStack** in a simulated AWS environment.
- Built a production ETL pipeline using **AWS Glue** and **Apache Airflow** to deliver curated, daily refreshed datasets to the Data Science team.

Exusia

December 2022 – July 2023

Data Engineer

Hyderabad, India

- Collaborated with the data governance team to build data lineage documentation and upload structured metadata to **Collibra**, improving data discoverability across the organisation.
- Developed **Python automation scripts** scheduled via **cron jobs** to verify reports and dispatch email alerts, reducing manual comparison effort by **up to 90%**.

Tata Consultancy Services (TCS)

October 2020 – December 2022

Software Engineer (Data)

Hyderabad, India

- As a fresher, was part of a team to develop a **batch ETL pipeline** using **Spark (on-prem)**, **Scala**, **Hive**, and **Shell Script** for a UK BFSI client to calculate refundable amounts to customers in cases of bank insolvency.

- Wrote **Data Quality** check scripts in Spark for finetuning the incoming data.
- Optimised a critical job, reducing runtime from **6 hours to 4 hours** — a **33% efficiency improvement** — through query tuning and pipeline restructuring.
- Built an **automation testing framework** that increased the QA team's efficiency by **40%**; maintained full project documentation across the delivery lifecycle.
- Managed source code with **Git/GitHub** and implemented **CI/CD pipelines** via **Jenkins** in an Agile delivery model.

PERSONAL PROJECTS

- **RAG Chatbot** – A comprehensive AI application that transcribes YouTube videos and local audio files using **OpenAI Whisper**, then enables intelligent Q&A through a **Retrieval-Augmented Generation (RAG)** chatbot powered by **Claude AI**.
- **Meal Suggestion** - A small **Agentic AI** project designed to automate daily meal planning. Unlike simple random generators, this agent **maintains state/memory** of the previous meals to ensure variety. It leverages a **local LLM (DeepSeek via Ollama)** to generate meal plans including macros, and sends to whatsapp using **Twilio** on a daily basis(scheduled via cron job).

CERTIFICATIONS

- **AZ-900 – Microsoft Azure Fundamentals** | Score: 820 / 900
- **DP-900 – Microsoft Azure Data Fundamentals** | Score: 823 / 900

EDUCATION

Techno Main Salt Lake

B.Tech in Electronics and Communication Engineering (ECE)

CGPA: **8.14 / 10**

2016 – 2020

Kolkata, India